

PH23C01 ENGINEERING PHYSICS

Anna University | Important Questions Analysis 2026

ASSESSMENT-PAPER BASED IMPORTANT QUESTIONS

UNIT I -- CRYSTAL STRUCTURE, BONDING & DEFECTS

2-Mark Questions

Q1. State differences between ionic and covalent bonding.

Probability: ★★★★★ Direct PYQ Assess-I Q1

Q2. Define Lattice, Basis and Unit Cell.

Probability: ★★★★★ Direct PYQ Assess-I Q3

Q3. Find Miller indices of plane with intercepts 2a, 3b, 2c.

Probability: ★★★★★ Direct PYQ Numerical

Q4. Define atomic packing factor.

Probability: ★★★★★ Anna Univ standard

Q5. What is a point defect? Name its types.

Probability: ★★★★★ Direct PYQ Assess-I

Q6. Distinguish Schottky and Frenkel defects.

Probability: ★★★★★ Anna Univ standard

Q7. State principle of Czochralski method.

Probability: ★★★★★ Direct PYQ Assess-I Q8i

Q8. State Bragg's law. Find max order for $d=2A$, wavelength= $1.5A$.

Probability: ★★★★★ Direct PYQ Assess-I Q8ii

13-Mark Questions

Q1 (13M): Derive APF for BCC (68%) and FCC (74%) with complete steps. (7M) + Numerical: find lattice constant and density given atomic radius. (6M)

Probability: ★★★★★ Direct PYQ Assess-I Q6(a)

Q2 (13M): Explain crystal defects -- point defects (vacancy, interstitial, Frenkel, Schottky) and line defects (edge, screw dislocation). Diagrams, examples, effects. (8M) + Schottky defect numerical $N \cdot \exp(-E/2kT)$. (5M)

Probability: ★★★★★ Direct PYQ Assess-I Q6(b)

15-Mark Questions

Q1 (15M): Czochralski method -- principle, setup, procedure, crystal quality, advantages and applications. (10M) + Bragg's law numerical: $d=2.0A$, $\lambda=1.5A$, find max diffraction order (Answer: $n_{\text{max}}=2$). (4M)

Probability: ★★★★★ Direct PYQ Assess-I Q8 -- VERY HIGH

UNIT II -- ELASTICITY & MECHANICAL PROPERTIES

2-Mark Questions

Q1. Define rigid body and conditions for rigidity.

Probability: ★★★★★ Direct PYQ Assess-I Q4

Q2. Two shafts with same torque -- which twists more and why?

Probability: ★★★★★ Direct PYQ Assess-I Q5

Q3. Write expression for depression at free end of cantilever.

Probability: ★★★★★ Direct PYQ

Q4. Compare square and circular cross-section beams for stiffness.

Probability: ★★★★★ Direct PYQ Assess-I Q7b

Q5. Define Poisson's ratio.

Probability: ★★★★★ Standard

Q6. State Hooke's law.

Probability: ★★★★★ Standard

Q7. Define bending moment and neutral axis.

Probability: ★★★★★ Standard

13-Mark Questions

Q1 (13M): Elastic behaviour of ductile and brittle materials -- stress-strain curves with all points (proportional limit, elastic limit, yield point, UTS, fracture point). Compare characteristics. (10M) + Torsional couple numerical. (3M)

Probability: ★★★★★ Direct PYQ Assess-I Q7(a)

Q2 (13M): Derive depression at free end of cantilever $y = WL^3 / 3EI$ using bending equation $M/I = E/R$. (10M) + Compare square vs circular beam depressions -- show ratio $y_{\text{square}}:y_{\text{circle}} = 3:\pi$. (3M)

Probability: ★★★★★ Direct PYQ Assess-I Q7(b)

UNIT III -- OSCILLATIONS, WAVES & ULTRASONICS

2-Mark Questions

Q1. Torsion pendulum -- if wire radius is halved, what happens to period? (Ans: 4T)

Probability: ★★★★★ Direct PYQ Assess-II Q1

Q2. Amplitude drops from 5 cm to 2 cm in 2 s. Find damping coefficient beta.

Probability: ★★★★★ Direct PYQ Assess-II Q2

Q3. What are expansion joints? Explain their function.

Probability: ★★★★★ Direct PYQ Assess-II Q3

Q4. State Doppler effect. Write the general frequency formula.

Probability: ★★★★★ PYQ numerical support

Q5. Distinguish A-scan and B-scan in medical ultrasonics.

Probability: ★★★★★ Direct PYQ Assess-II

Q6. List two applications of ultrasonics in medicine.

Probability: ★★★★★

13-Mark Questions

Q1 (13M): Torsion pendulum -- derive expressions for moment of inertia of disc ($I = MR^2/2$) and rigidity modulus of wire. Complete theory with diagrams. (10M) + Numerical: Given T_1 (disc alone) and T_2 (extra mass), find I and G . (3M)

Probability: ★★★★★ Direct PYQ Assess-II Q6(a)

Q2 (13M): Describe A-scan (4M), B-scan (4M) and TM-scan (5M) methods for medical imaging using ultrasonic waves. Draw neat diagrams for each method.

Probability: ★★★★★ Direct PYQ Assess-II Q6(b)

Q3 (13M): Doppler effect -- derive formula for all cases (source moving, observer moving, both moving). Applications in medicine. (10M) + Numerical: Car A at 20 m/s (450 Hz), Car B at 35 m/s observer. Find f_{obs} . [Ans: ~469 Hz] (3M)

Probability: ★★★★★ Direct PYQ Assess-II Q8

15-Mark Questions

Q1 (15M): Michelson interferometer -- construction, working, fringe formation, applications. (10M) + Doppler numerical (same as Q3 above). (4M)

Probability: ★★★★★ Direct PYQ Assess-II Q8 -- VERY HIGH

UNIT IV -- OPTICS, LASERS & FIBER OPTICS

2-Mark Questions

Q1. What is coherence? Distinguish temporal and spatial coherence.

Probability: ★★★★★ Direct PYQ Assess-II Q4

Q2. Air-wedge filled with liquid (n) -- what happens to fringe spacing?

Probability: ★★★★★ Direct PYQ Assess-II Q7b(ii)

Q3. Distinguish active and passive fiber optic sensors.

Probability: ★★★★★ Direct PYQ Assess-II Q5

Q4. Why must stimulated emission be introduced as fundamental radiation process?

Probability: ★★★★★ Direct PYQ Assess-II Q7a(ii)

Q5. Define numerical aperture and acceptance angle.

Probability: ★★★★★ Very frequent

Q6. What is population inversion?

Probability: ★★★★★ Essential laser concept

Q7. State two advantages of Nd:YAG laser over ruby laser.

Probability: ★★★★★ Assess-II Q7a support

13-Mark Questions

Q1 (13M): Nd:YAG laser -- construction, working, energy level diagram (4-level system), applications. (10M) + Short note: Why must stimulated emission be introduced as a fundamental radiation process? (3M)

Probability: ★★★★★ Direct PYQ Assess-II Q7(a)

Q2 (13M): Derive numerical aperture and acceptance angle for step-index optical fibre from first principles (Snell's law, TIR). (10M) + Numerical: $n_1=1.5$, $n_2=1.45$ -- find NA and acceptance angle. (3M)

Probability: ★★★★★ Direct PYQ Assess-II Q7(b)(i)

Q3 (13M): Air wedge experiment -- interference theory, fringe width derivation $\beta = \lambda/2\theta$, measurement of wire diameter, applications. (10M) + Numerical: 20 fringes per cm, $\lambda=5893\text{\AA}$ -- find wire thickness. (3M)

Probability: ★★★★★ Anna Univ Very High

15-Mark Questions

Q1 (15M): Derive NA and acceptance angle. Explain fiber optic communication system with complete block diagram (transmitter, fiber, repeater, receiver). Describe fiber sensors (active/passive). (12M) + Numerical: $n_1=1.50$, $n_2=1.46$ -- find NA, acceptance angle, critical angle. (3M)

Probability: ★★★★★ Very High -- Part C

UNIT V -- CRYSTAL GROWTH & X-RAY DIFFRACTION

2-Mark Questions

Q1. State principle of Czochralski method of crystal growth.

Probability: ★★★★★ Direct PYQ Assess-I Q8(i)

Q2. State Bragg's law. Find max order for $d=2A$, wavelength= $1.5A$.

Probability: ★★★★★ Direct PYQ Assess-I Q8(ii)

Q3. What is the Bridgman method of crystal growth?

Probability: ★★★★★ Alternate PYQ

Q4. What is the powder diffraction method? State one application.

Probability: ★★★★★ Anna Univ

15-Mark Questions

Q1 (15M): Czochralski method -- principle, experimental setup, procedure, crystal quality control, advantages and applications. (10M) + Bragg's law numerical: $d=2.0A$, $\lambda=1.5A$ -- find maximum order of diffraction. [Answer: $n_{\text{max}} = 2$] (4M)

Probability: ★★★★★ Direct PYQ Assess-I Q8 -- MOST PROBABLE PART C

FINAL PRIORITY TABLE -- DIRECT PYQ EVIDENCE

Unit	Topic	Stars	Evidence
I	BCC/FCC Packing Factor (APF)	★★★★★	Direct Assess-I Q6a
I	Crystal Defects (Point & Line)	★★★★★	Direct Assess-I Q6b
I	Miller Indices + Numerical	★★★★★	Direct PYQ Numerical
II	Stress-Strain Diagram (Ductile/Brittle)	★★★★★	Direct Assess-I Q7a
II	Cantilever Depression Formula	★★★★★	Direct Assess-I Q7b
II	Square vs Circular Beam (3:pi)	★★★★★	Direct Assess-I Q7b
III	Torsion Pendulum (I & G)	★★★★★	Direct Assess-II Q6a
III	A-scan, B-scan, TM-scan	★★★★★	Direct Assess-II Q6b
III	Doppler Effect + Numerical	★★★★★	Direct Assess-II Q8
IV	Nd:YAG Laser	★★★★★	Direct Assess-II Q7a
IV	Optical Fiber NA & Acceptance Angle	★★★★★	Direct Assess-II Q7b(i)
IV	Michelson Interferometer	★★★★★	Direct Assess-II Q8
IV	Air Wedge Experiment	★★★★★	Anna Univ Very High
V	Czochralski Method	★★★★★	Direct Assess-I Q8i
V	Bragg's Law & Max Order	★★★★★	Direct Assess-I Q8ii
II	Torsion of Shafts / Rigidity Modulus	★★★★■	PYQ Numerical
IV	Coherence (Temporal/Spatial)	★★★★■	Direct Assess-II Q4
IV	Air-Wedge with Liquid (fringe change)	★★★★■	Direct Assess-II Q7b(ii)
III	Damped Oscillations / beta calculation	★★★★■	Direct Assess-II Q2
I	Ionic/Covalent Bonding	★★★★■	Direct Assess-I Q1

Prepare ALL 5-star topics first -- they are from ACTUAL assessment papers!